

**Beautiful Dua
from Quran**

رَبِّ اشْرَحْ لِي صَدْرِي وَيَسِّرْ لِي أَمْرِي
وَاحْلُلْ عُقْدَةً مِّن لِّسَانِي يَفْقَهُوا قَوْلِي

Punjab University

Bs (4 Years) Chemistry

Semester Sixth

Solve Past paper

(2022)

Q. What is CULLET?

Repeat 2023

Q. Write any two application of hydrogenation?

Repeat 2023

Q. Define UNIT OPERATION with example?

Repeat 2023

Q. What is CUMENE HYDROPEROXIDE?

Repeat 2023

Q. Draw the structure of PHTHALIC ANHYDRIDE?

Repeat 2023

Q. Which one ACID is produced by oxidation of carbohydrates?

Repeat 2023

Q. What is SAPONIFICATION?

Repeat 2023

Q. Enlist any two oxidizing agents?

An oxidizing agent (or oxidant) is a substance that gains electrons in a chemical reaction and causes another substance to lose electrons. In other words, it **oxidizes** another substance by accepting electrons, while itself gets **reduced**.

Two oxidizing agents with their chemical reactions:

1. Potassium Permanganate (KMnO_4)

- Reaction in acidic medium:**

Potassium permanganate oxidizes iron(II) sulfate (FeSO_4) to iron(III) sulfate ($\text{Fe}_2(\text{SO}_4)_3$) in the presence of sulfuric acid (H_2SO_4).

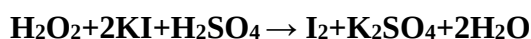


Here, KMnO_4 (oxidizing agent) is reduced to MnSO_4 , while FeSO_4 is oxidized to $\text{Fe}_2(\text{SO}_4)_3$.

2. Hydrogen Peroxide (H_2O_2)

- Reaction with potassium iodide (KI):**

Hydrogen peroxide oxidizes potassium iodide to iodine (I_2) in an acidic medium.



Here, H_2O_2 (oxidizing agent) is reduced to H_2O , while KI is oxidized to I_2 .

Haq Nawaz Bhatti Page Number 26

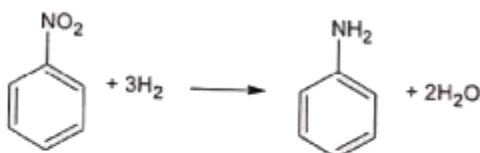
Q. Mention the major application of NITRATION of benzene?

1.2.3.3 Applications

Some potential applications of nitration are enlisted as under

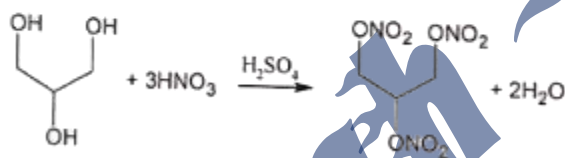
i) Reduction to Amines

Nitrobenzene is reduced to aniline in the stream of hydrogen at 350°C passing over reduced copper ammonium nitrate on silica gel.



ii) Nitroglycerine

Nitration of glycerol with weak acid results in the formation of organic-inorganic ester. It is very sensitive to stock render it of little as a high explosive. A mixture of nitroglycerine and siliceous earth is not sensitive to shock, and this mixture is useful high explosive and dynamite.



iii) Nitrocellulose

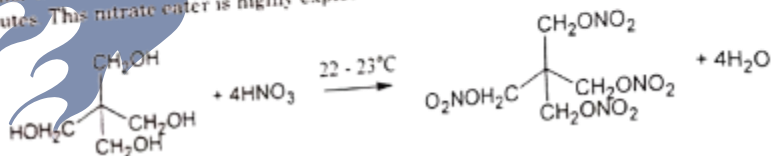
Nitration of cellulose is conducted with mixed acid (sulphuric acid 60%, nitric acid 25-28%, water 9-11%) results in nitrate ester to 13%. The rate of explosive decomposition

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for this material is much slower than that of nitroglycerine so it may be employed as a propeller. It is the basis of smokeless powder for gun cotton.

iv) Pentaerythritol

Pentaerythritol is nitrated by nitric acid alone to pentaerythritol tetranitrate. The alcohol is mixed with 95% nitric acid and 18°C and then brought to 22 to 23°C for 20 minutes. This nitrate ester is highly explosive.



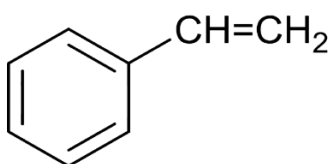
Q. Write down the IUPAC name of STYRENE?

The IUPAC name of styrene is:

Phenylethene

This is because styrene consists of a **phenyl group** (C_6H_5 -) attached to an **ethene (ethylene, -CH=CH₂)** group.

Its structure is:



Q. What is "GLASS"?

Glass is an **amorphous (non-crystalline) solid material** that is typically transparent or translucent and is made by cooling molten materials such as silica (SiO_2) rapidly to prevent crystallization. It is a versatile material with a wide range of applications due to its unique properties.

Composition of Glass:

The most common type of glass is **soda-lime glass**, which is composed of:

- **Silica (SiO_2)** - The main component (about 70-75%).
- **Sodium oxide (Na_2O)** - Lowers the melting point of silica.
- **Calcium oxide (CaO)** - Provides stability and durability.

Other types of glass include:

- **Borosilicate glass** (e.g., Pyrex) - Contains boron oxide (B_2O_3) for thermal and chemical resistance.
- **Lead glass** - Contains lead oxide (PbO) for higher refractive index and brilliance (used in crystal glassware).

Haq Nawaz Bhatti Page Number 126

Q. What is percentage of Silica in HIGH SILICA glass?

- **High silica glass**, also known as **fused quartz**, **fused-silica glass**, or **vitreous-silica glass**, contains **99% or more silica (SiO_2)**.
- It is prepared by the **pyrolysis of silicon tetrachloride** or **pure sand** at high temperatures.

- This glass has **very low thermal expansion**, a **high softening point**, and is **very hard, heat-resistant, and weathering-resistant**.
- It is **extraordinarily transparent to ultraviolet (UV) radiation**.
- High silica glass is used for **high-temperature applications**, such as **furnace tubes, lighting tubes, and melting crucibles**.

Haq Nawaz Bhatti Page Number 133

Q. Enlist major ingredients of SODA GLASS?

The major ingredients of **Soda-Lime Glass** (also called **window glass**) as per the image are:

1. **Silica (SiO_2): 70-74%** - The primary component that forms the glass structure.
2. **Sodium Oxide (Na_2O): 13-18%** - Lowers the melting point of silica, making the glass easier to work with.
3. **Lime (Calcium Oxide, CaO): 8-13%** - Stabilizes the glass and improves its durability.

Additional

- This type of glass is **transparent, easily formed**, and widely used for **windows, low-temperature incandescent light bulbs, and tableware**.
- It has **high thermal expansion** and **poor resistance to heat**.
- **Container glass** (used for bottles and jars) is a variation of soda-lime glass with **more alumina and calcium**, and **less sodium and magnesium**, making it more resistant to water erosion.

Haq Nawaz Bhatti Page Number 133

Q. What is ANNEALING?

Annealing is a process used in glass manufacturing to cool newly shaped glass articles slowly. This process is essential to prevent the glass from becoming brittle due to internal strains caused by rapid cooling. Here's how it works based on the image content:

1. **Purpose of Annealing:**
 - Annealing allows the glass molecules to arrange themselves in a way that eliminates internal strain as the glass cools.
 - Without annealing, the glass would be brittle and prone to breaking due to internal stresses.
2. **Process of Annealing:**

- The glass articles are cooled slowly in a tunnel-like oven called a **lehr**, which is **50 to 60 feet long**.
- The temperature at one end of the lehr is slightly below the softening point of glass (**500-600°C**), and it gradually decreases along the length of the oven.
- At the other end, the temperature is nearly the same as room temperature.
- The newly shaped glass articles are placed on a moving belt and travel from the hotter end to the cooler end over a few hours.

3. Importance of Slow Cooling:

- Slow cooling ensures that the glass remains strong and free from internal stresses.
- High-quality glasses may require longer annealing times to achieve the desired properties.

Haq Nawaz Bhatti Page Number 131

Q. Why LAMINATED safety glass does not break easily?

Laminated safety glass does not break easily because of its unique construction and the materials used.

Layered Structure:

- Laminated glass is made by bonding **two or more layers of safety glass** (often annealed glass) with an adhesive, typically **polyvinyl butyral (PVB)**.
- These layers are held together by **plastic interlayers**, which add strength and flexibility.

2. Impact Resistance:

- When laminated glass is struck, the glass layers may crack, but the **plastic interlayer** holds the broken pieces together.
- This prevents the glass from shattering into sharp, dangerous shards.

3. Strength and Safety:

- The number of glass layers and plastic interlayers can be adjusted to increase or decrease the strength of the glass.
- This makes laminated glass highly resistant to breaking and ideal for applications where safety is critical, such as **automobile windshields** and **bullet-resistant glass**.

4. Applications:

- Laminated glass is used in **automobile windshields** because it provides strength and prevents shattering during accidents.

- It is also used in **bullet-resistant glass**, which is constructed using multiple layers of laminated glass for added protection.

Haq Nawaz Bhatti Page Number 134

Long Questions

Q. Briefly describe the role of Borax and Soda in the manufacturing of glass?

Repeat 2023

Q. Describe the classification of surfactants with examples?

Repeat 2023

Q. Describe the significance and mechanism of Sulphonation of benzene?

Haq Nawaz Bhatti Page Number 13

Q. Explain the chemical changes involved in manufacturing of Oxalic acid from Sodium formate?

Haq Nawaz Bhatti Page Number 49

Q. Write a note on chemically strengthened SAFETY GLASS?

Chemically Strengthened Safety Glass is a type of glass that undergoes a chemical process to enhance its strength and durability. Unlike tempered glass, which is strengthened by thermal treatment, chemically strengthened glass is treated through an ion-exchange process. Here's a detailed note on it:

Process of Chemical Strengthening:

1. Ion Exchange:

- The glass is submerged in a **molten salt bath** (typically potassium nitrate, KNO_3) at high temperatures (around $300\text{--}400^\circ\text{C}$).
- Smaller sodium ions (Na^+) in the glass surface are replaced by larger potassium ions (K^+) from the salt bath.
- This ion exchange creates **compressive stress** on the surface of the glass, making it stronger.

2. Surface Compression:

- The larger potassium ions create a "squeezing" effect on the glass surface, resulting in high compressive stress.
- This compression makes the glass highly resistant to scratches, cracks, and impacts.

Properties of Chemically Strengthened Glass:

1. High Strength:

- It is significantly stronger than regular glass and can withstand greater mechanical stress.
- The strength can be 6–8 times higher than untreated glass.

2. Thin and Lightweight:

- Chemically strengthened glass can be made thin while maintaining high strength, making it ideal for modern devices.

3. No Surface Distortion:

- Unlike tempered glass, chemically strengthened glass does not warp or distort, making it suitable for optical applications.

4. Breakage Pattern:

- When it breaks, it forms smaller, less dangerous pieces compared to regular glass, though not as small as tempered glass.

Applications:

1. Consumer Electronics:

- Used in smartphone screens, tablets, and wearable devices due to its scratch resistance and strength.

2. Aviation and Automotive:

- Used in aircraft windows and car windshields for its lightweight and high strength.

3. Optical and Medical Devices:

- Ideal for lenses, microscope slides, and medical equipment due to its clarity and durability.

4. Architectural Glass:

- Used in high-strength windows and glass facades.

Advantages Over Tempered Glass:

- Can be cut or drilled after strengthening (unlike tempered glass, which must be cut before tempering).
- No risk of spontaneous breakage (a issue sometimes seen in tempered glass).
- Suitable for thinner glass applications.

Haq Nawaz Bhatti Page Number 134

Diikash

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Punjab University

Bs (4 Years) Chemistry

Semester Sixth

Solve Past paper

(2021)



Q. Write down the chemical reactions involved in manufacturing of OXALIC ACID from SODIUM FORMATE?

Repeat in 2022 long

Q. Differentiate between UNIT OPERATION and UNIT PROCESS?

Difference Between Unit Operation and Unit Process

Unit Operation and Unit Process are **fundamental concepts in chemical engineering** that describe different steps in industrial processing.

Aspect	Unit Operation	Unit Process
Definition	A Unit Operation refers to a physical step in a process that does not involve any chemical change in the substance.	A Unit Process refers to a chemical step in a process where a chemical reaction occurs, forming a new substance .
Nature of Change	A Unit Operation involves a physical change , such as separation or mixing, where the substance remains the same.	A Unit Process involves a chemical change , where the substance undergoes a chemical reaction to form a new product.
Examples	Common examples of Unit Operations include distillation, filtration, drying, crystallization, and evaporation , where physical separation or transformation occurs.	Common examples of Unit Processes include nitration, oxidation, polymerization, esterification, and hydrolysis , where new chemical compounds are formed.
Purpose	The purpose of a Unit Operation is to separate, purify, concentrate, or change the physical state of a material.	The purpose of a Unit Process is to convert raw materials into desired chemical products through chemical reactions.
Equipment Used	Unit Operations use physical equipment such as filters, dryers, distillation columns, and evaporators .	Unit Processes use chemical reactors, catalysts, and reaction-specific equipment to facilitate chemical transformations.
Industry Application	Unit Operations are used in various industries , including food, pharmaceuticals, and chemical manufacturing .	Unit Processes are commonly used in chemical, petrochemical, and pharmaceutical industries , where chemical reactions are required.

Q. Describe the objective of GLASS ANNEALING?

Repeat in 2022

Q. Differentiate between SOAP and DETERGENT?

Difference Between Soap and Detergent

Soaps and detergents are both cleaning agents, but they have different chemical compositions and properties.

Aspect	Soap	Detergent
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Definition	Soap is a natural cleaning agent made by reacting fats or oils with an alkali (like NaOH or KOH).	Detergent is a synthetic cleaning agent made from petrochemicals or synthetic surfactants.
Chemical Composition	Soaps are sodium or potassium salts of fatty acids.	Detergents are made from sulfonates or sulfates (e.g., sodium lauryl sulfate).
Formation Reaction	Saponification Reaction: $\text{Fat/Oil} + \text{NaOH} \rightarrow \text{Soap} + \text{Glycerol}$ Example: $\text{C}_3\text{H}_5(\text{C}_{17}\text{H}_{35}\text{COO})_3 + 3\text{NaOH} \rightarrow 3\text{C}_{17}\text{H}_{35}\text{COONa} + \text{C}_3\text{H}_5(\text{OH})_3$	Detergent Formation: $\text{Alkylbenzene} + \text{H}_2\text{SO}_4 \rightarrow \text{Alkylbenzene Sulfonic Acid}$ $\text{Alkylbenzene Sulfonic Acid} + \text{NaOH} \rightarrow \text{Sodium Alkylbenzene Sulfonate (Detergent)}$
Solubility in Hard Water	Soaps form scum in hard water due to reaction with calcium and magnesium ions.	Detergents do not form scum and work well in both soft and hard water.
Foaming Ability	Soaps do not lather well in hard water.	Detergents lather well even in hard water.
Biodegradability	Soaps are biodegradable and do not harm the environment.	Some detergents are non-biodegradable, leading to environmental pollution.
Uses	Used for bathing, handwashing, and mild cleaning.	Used for laundry, dishwashing, and industrial cleaning.

Haq Nawaz Bhatti Page Number 149 and 165

Q. Write any two application of HYDROGENATION?

Repeat in 2022

Q. What is primary composition of SODA-LIME GLASS?

Repeat in 2022

Q. Mention the catalyst used in manufacturing of PHTHALIC ANHYDRIDE?

Repeat in 2023

Q. What is CULLET in glass manufacturing?

Repeat in 2023

Q. List down materials required for manufacturing of ETHYL ACETATE?

Materials Required for Manufacturing of Ethyl Acetate

The production of Ethyl Acetate ($\text{CH}_3\text{COOCH}_2\text{CH}_3$) is commonly done via Esterification of Ethanol and Acetic Acid. The required materials include:

1. Raw Materials:

- Ethanol ($\text{C}_2\text{H}_5\text{OH}$) – Acts as the alcohol reactant.
- Acetic Acid (CH_3COOH) – Acts as the carboxylic acid reactant.

2. Catalysts:

- Sulfuric Acid (H_2SO_4) – Used as a catalyst to speed up the esterification reaction.
- p-Toluenesulfonic Acid (p-TSA) – Sometimes used as an alternative catalyst.

3. Dehydrating Agent (Optional):

- Molecular Sieves – Used to remove water and shift equilibrium towards ester formation.

4. Solvents (Optional):

- Sometimes organic solvents like benzene or toluene are used in azeotropic distillation to remove water.

Reaction:



Q. What are cationic and anionic surfactants?

Repeat in 2023 Long

Long Questions

Q. Explain principle and significance of DISTILLATION of DISTILLATION with example?

Haq Nawaz Bhatti Page Number 69

Q. How COLOR is imparted in glass. Explain all possible phenomenon?

Color in glass is imparted through various mechanisms, primarily involving the addition of **metal ions**, **colloidal particles**, or **nanoparticles** that interact with light. These mechanisms alter the way light is absorbed, transmitted, or scattered, resulting in different colors.

1. Reduction-Oxidation (Redox) Reactions:

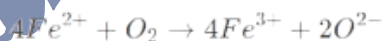
- **Phenomenon:**

The oxidation state of metal ions in the glass changes during the melting process leading to different colors.

- **Example:**

- **Iron (Fe) in Glass:**

- **Fe²⁺ (Ferrous Ion):** Imparts a **bluish-green color**.
 - **Fe³⁺ (Ferric Ion):** Imparts a **yellowish color**.
 - The balance between Fe²⁺ and Fe³⁺ depends on the **oxygen fugacity** (oxygen availability) in the melt.
 - Reaction:



2. Formation of Colloidal Particles:

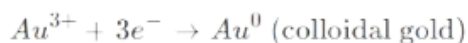
- **Phenomenon:**

Metal ions are reduced to form **colloidal particles** within the glass matrix, which scatter light and impart color.

- **Example:**

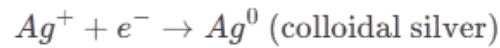
- **Gold (Au) Colloids:**

- Gold chloride (AuCl₃) is added to the glass melt.
 - During cooling, gold ions are reduced to form **colloidal gold particles**, which impart a **ruby red color**.
 - Reaction:



- **Silver (Ag) Colloids:**

- Silver nitrate (AgNO_3) is added to the glass melt.
- Silver ions are reduced to form **colloidal silver particles**, which impart a **yellow color**.
- Reaction:



3. Formation of Semiconductor Nanoparticles:

- **Phenomenon:**

Metal ions react with sulfur or selenium to form **semiconductor nanoparticles** (e.g., CdS , CdSe), which impart color due to quantum confinement effects.

- **Example:**

- **Cadmium Sulfide (CdS) Nanoparticles:**

- Cadmium oxide (CdO) and sulfur (S) are added to the glass melt.
- They react to form **CdS nanoparticles**, which impart a **yellow color**.
- Reaction:



- **Cadmium Selenide (CdSe) Nanoparticles:**

- Cadmium oxide (CdO) and selenium (Se) are added to the glass melt.
- They react to form **CdSe nanoparticles**, which impart a **red color**.

- **Reaction:**



4. Formation of Color Centers (Radiation-Induced Coloration):

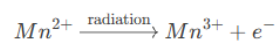
- **Phenomenon:**

High-energy radiation (e.g., UV, X-rays, or gamma rays) creates **defects** in the glass structure, which absorb specific wavelengths of light and impart color.

- **Example:**

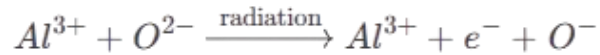
- **Amethyst Glass:**

- Manganese (Mn^{2+}) is added to the glass.
- When exposed to radiation, Mn^{2+} is oxidized to Mn^{3+} , creating **color centers** that impart a **purple color**.
- Reaction:



- **Smoky Glass:**

- Aluminum (Al^{3+}) and oxygen defects are present in the glass.
- Radiation creates **electron-hole pairs**, which impart a **brown or gray color**.
- Reaction:



5. Formation of Metal Oxides (Dichroic Glass):

- **Phenomenon:**

Thin layers of metal oxides are deposited on the glass surface, creating **interference effects** that impart color.

- **Example:**

- **Titanium Oxide (TiO_2):**

- Titanium tetrachloride ($TiCl_4$) is vaporized and reacts with oxygen to form a thin layer of TiO_2 on the glass surface.
 - Reaction:



- The TiO_2 layer creates **interference effects**, imparting a **multicolored appearance**.

6. Formation of Opacifiers and Pigments:

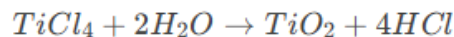
- **Phenomenon:**

Metal oxides or sulfides are added to the glass to scatter or absorb light, imparting color.

- **Example:**

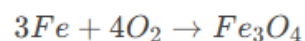
- **Titanium Dioxide (TiO_2):**

- Added to the glass to create a **white, opaque color**.
 - Reaction:



- **Iron Oxide (Fe_3O_4):**

- Added to the glass to create a **black color**.
 - Reaction:



Q. Briefly describe the manufacturing of Phenol from CUMENE process?

Repeat in 2023 Short question

Q. Write a note on RECOVERY OF GLYCERIN in soap manufacturing?

Haq Nawaz Bhatti Page Number 151

Q. Briefly describe the CLEANSING MECHANISM of soap?

Haq Nawaz Bhatti Page Number 161

Ditkash

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Punjab University

Bs (4 Years) Chemistry

Semester Sixth

Solve Past paper

(2020)

Q. Write down the chemical reaction for preparation of phthalic anhydride from naphthalene?

Repeat in 2023

Q. Describe the annealing of glass?

Repeat in 2023

Q. Give two industrial applications of HYDRATION?

Repeat in 2023

Q. Draw a flow sheet diagram of oxalic acid manufacturing from Sodium formate?

Repeat in 2023

Q. Define UNIT CHEMICAL PROCESSES with examples?

Repeat in 2023

Q. What is significance of Safety glass?

Safety glass is designed to minimize injury risks in case of breakage, making it essential in various applications.

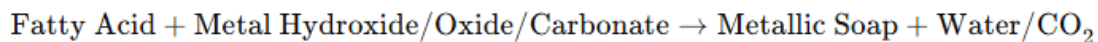
1. **Reduced Injury Risk:** Unlike regular glass, **tempered glass** shatters into small, blunt pieces, reducing the risk of severe cuts. **Laminated glass**, with its interlayer, holds fragments together, preventing sharp shards from scattering.
2. **Enhanced Security:** **Laminated safety glass** is commonly used in **car windshields and buildings**. Its strong interlayer resists penetration, making forced entry more difficult.
3. **Impact Resistance:** Safety glass is **stronger than regular glass**, making it ideal for **vehicle windows, shower doors, and glass railings**, where breakage risks are high.
4. **Sound Insulation:** **Laminated safety glass** also provides **soundproofing**, making it useful in **offices, homes, and urban areas** to reduce noise pollution.
5. **UV Protection:** Some types of **laminated glass** can block harmful **ultraviolet (UV) rays**, protecting **interiors, furniture, and occupants** from sun damage.
6. **Compliance with Safety Standards:** Many **building codes and automotive regulations** mandate the use of safety glass to **ensure public safety**.

Q. What are Metallic soaps?

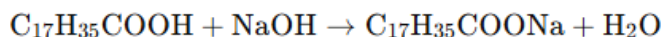
Metallic soaps are salts of fatty acids in which metal ions replace the hydrogen ion of the carboxyl (-COOH) group. These soaps are typically formed by reacting **metal oxides, hydroxides, or carbonates** with long-chain **carboxylic acids** (fatty acids). Unlike regular soaps (which are soluble in water), most metallic soaps are **insoluble in water** and are used in **industrial applications** such as lubricants, stabilizers, waterproofing agents, and plastic additives.

Chemical Reaction for Metallic Soap Formation

The general reaction for the formation of metallic soaps is:



Example: Formation of Sodium Stearate (Sodium Soap)

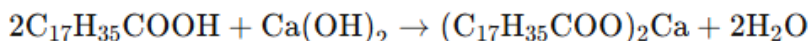


Here:

- $\text{C}_{17}\text{H}_{35}\text{COOH}$ = **Stearic Acid** (Fatty Acid)
- NaOH = **Sodium Hydroxide** (Alkali)
- $\text{C}_{17}\text{H}_{35}\text{COONa}$ = **Sodium Stearate** (A Metallic Soap)
- H_2O = **Water**

For heavy metal soaps, similar reactions occur, but with calcium, magnesium, zinc, or aluminum compounds.

Example: Formation of Calcium Stearate (Water-Insoluble Soap)



Q. How evaporators are classified?

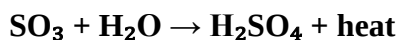
Haq Nawaz Bhatti Page Number 65

Why SO_3 is not directly added to water in the Contact Process?

In the Contact Process, SO_3 (sulfur trioxide) is not directly added to water because it reacts violently, forming a dense mist of sulfuric acid droplets instead of a smooth solution.

Reasons:

1. Highly Exothermic Reaction:



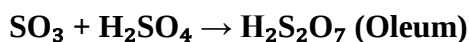
The reaction releases a large amount of heat, causing sulfuric acid to vaporize into a mist, making it difficult to handle.

2. Formation of Acid Mist:

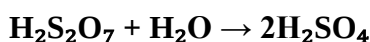
Instead of forming a liquid solution, SO_3 forms fine droplets of sulfuric acid that do not easily dissolve in water, making the process inefficient.

Solution Used in the Contact Process:

SO_3 is first absorbed in concentrated H_2SO_4 to form oleum ($\text{H}_2\text{S}_2\text{O}_7$).



Oleum is then diluted with water to obtain sulfuric acid of desired concentration.



This method prevents acid mist formation and ensures safe and efficient production of sulfuric acid.

Q. How GLYCERIN is recovered from spent LYE?

Repeat in Long

Long Questions

Q. Briefly describe the role of following raw materials for the manufacturing of glass?

a. Borax

b. Sand

Repeat

b) Describe manufacturing of PHTHALIC ANHYDRIDE in industry from naphthalene?

Repeat

c) What is the composition of laboratory glassware? How this composition is appropriate for laboratory work?

Laboratory glassware is typically made from **borosilicate glass**, which has a specific composition that makes it highly suitable for laboratory applications. The composition and its appropriateness for laboratory work are as follows:

Composition of Laboratory Glassware

Laboratory glassware is primarily made from **borosilicate glass**, which is specifically designed to meet the rigorous demands of laboratory environments. The composition of borosilicate glass is as follows:

1. **Silica (SiO_2):** 70-80%
 - Forms the primary network of the glass, providing structural integrity and thermal stability.
2. **Boron Oxide (B_2O_3):** 7-13%
 - Reduces the coefficient of thermal expansion, making the glass resistant to thermal shock.
3. **Alumina (Al_2O_3):** 2-7%
 - Enhances chemical durability and mechanical strength.
4. **Sodium Oxide (Na_2O) and Potassium Oxide (K_2O):** 4-8%
 - Act as fluxing agents to lower the melting point and improve workability during manufacturing.
5. **Trace Oxides:**
 - Small amounts of other oxides may be added to improve specific properties like clarity or resistance to certain chemicals.

Why Borosilicate Glass is Ideal for Laboratory Work

1. **Thermal Resistance**
 - Borosilicate glass has a low coefficient of thermal expansion, allowing it to withstand rapid temperature changes without cracking. This makes it suitable for applications involving heating, cooling, and autoclaving.
2. **Chemical Durability**
 - It is highly resistant to a wide range of chemicals, including acids, bases, and organic solvents. This ensures that the glassware does not react with or contaminate the substances being used.

3. Mechanical Strength

- The inclusion of alumina increases the mechanical strength of the glass, making it more durable and less prone to breakage during handling.

4. Transparency and Clarity

- The glass is transparent, allowing for easy observation of reactions, measurements, and processes inside the glassware.

5. Inertness

- Borosilicate glass is chemically inert, meaning it does not react with most substances. This ensures the purity and accuracy of experimental results.

6. Durability and Longevity

- The glass is highly durable and can withstand repeated use, sterilization, and cleaning, making it cost-effective for laboratory settings.

d) Describe SOAP manufacturing by KETTLE process?

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e) How NITRATION of benzene is carried out in industry?

Repeat in short

f) Write a note on COLORED glass?

Repeat

**Beautiful Dua
from Quran**

رَبِّ اشْرَحْ لِي صَدْرِي وَيَسِّرْ لِي أَمْرِي
وَاحْلُلْ عُقْدَةً مِّن لِّسَانِي يَفْقَهُوا قَوْلِي

Punjab University

Bs (4 Years) Chemistry

Semester Sixth

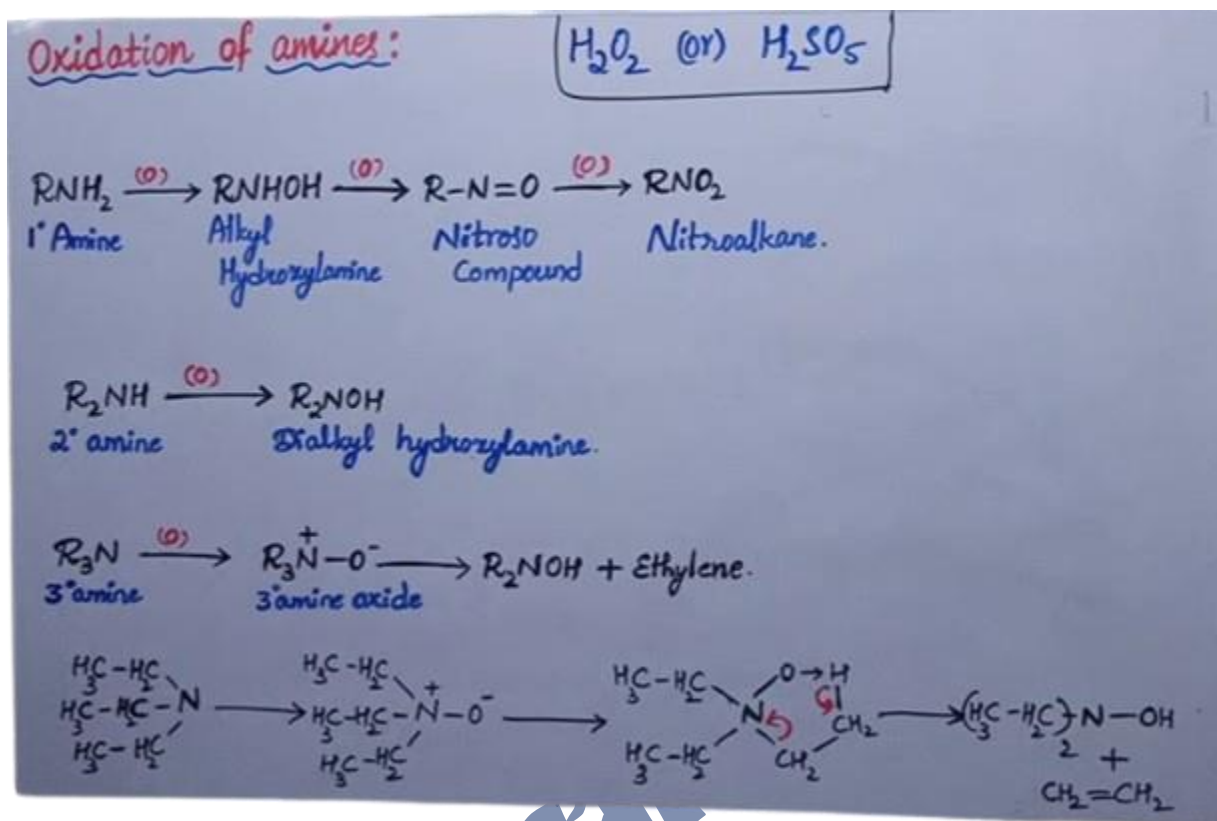
Solve Past paper

(2019)

Q. Define UNIT CHEMICAL PROCESSES with examples?

Repeat

Q. Discuss oxidation of AMINES?



Oxidation of Amines

Oxidation of amines depends on their type (primary, secondary, or tertiary). The reaction is carried out using oxidizing agents like H_2O_2 (hydrogen peroxide) or H_2SO_4 (sulfuric acid).

1. Oxidation of Primary Amines (R-NH_2)

- Forms **alkyl hydroxylamine (R-NHOH)**, which further oxidizes to **nitroso compounds (R-N=O)** and finally to **nitroalkanes (RNO_2)**.
- Example: An alkyl amine like ethylamine ($\text{C}_2\text{H}_5\text{NH}_2$) can be converted into nitroethane ($\text{C}_2\text{H}_5\text{NO}_2$).

2. Oxidation of Secondary Amines (R_2NH)

- Produces **dialkyl hydroxylamines ($\text{R}_2\text{N-OH}$)** as the main oxidation product.
- Example: Dimethylamine ($(\text{CH}_3)_2\text{NH}$) oxidizes to dimethylhydroxylamine ($(\text{CH}_3)_2\text{N-OH}$).

3. Oxidation of Tertiary Amines (R_3N)

- Forms **amine oxides** ($R_3N \rightarrow O$), which are used in surfactants and detergents.
- Example: Trimethylamine ($N(CH_3)_3$) oxidizes to trimethylamine oxide ($N(CH_3)_3 \rightarrow O$).

Reaction Mechanism Example

- The oxidation of **tertiary amines** can further break down into smaller compounds like **ethylene and hydroxylamines**, as shown in the diagram.
- This process is important in polymer chemistry and organic synthesis.

[https://chem.libretexts.org/Bookshelves/Organic_Chemistry/Basic Principles of Organic Chemistry \(Roberts and Caserio\)/23%3A Organonitrogen Compounds I - Amines/23.11%3A Oxidation of Amines](https://chem.libretexts.org/Bookshelves/Organic_Chemistry/Basic_Principles_of_Organic_Chemistry_(Roberts_and_Caserio)/23%3A_Organonitrogen_Compounds_I_-_Amines/23.11%3A_Oxidation_of_Amines)

Q. Describe the structure of CATIONIC surfactants?

Repeat

Q. Draw the flow sheet diagram of NITRATION process?

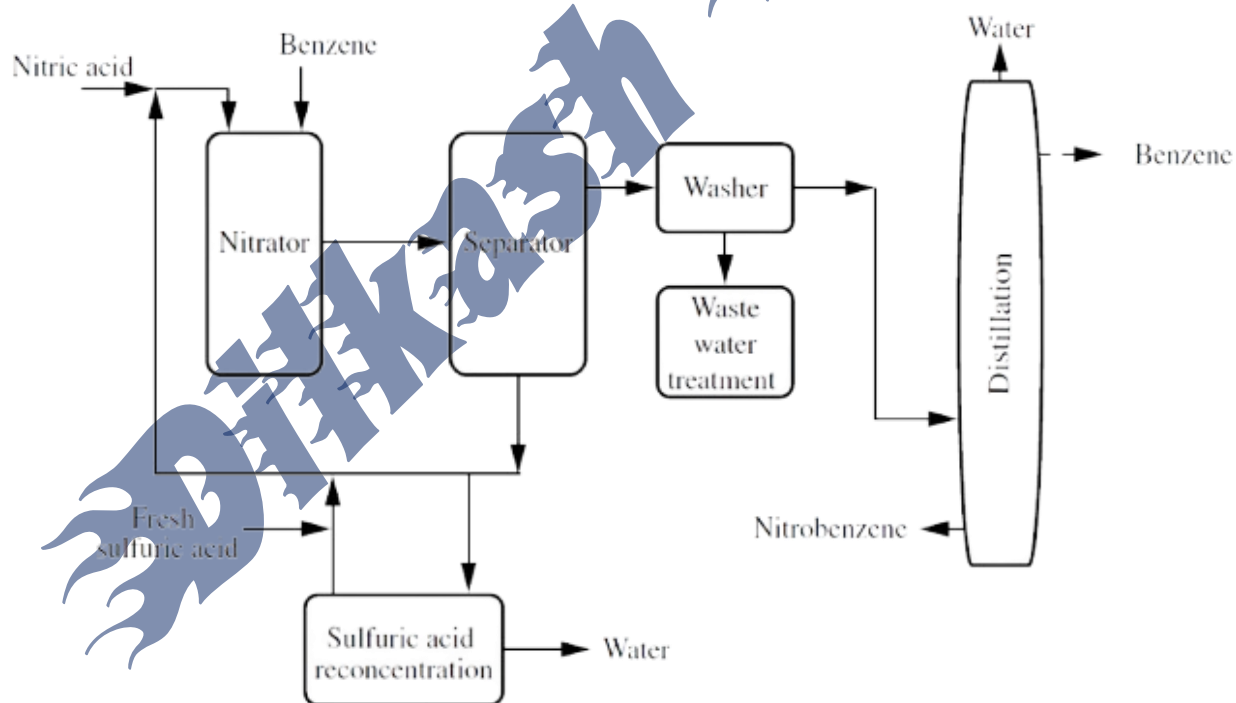
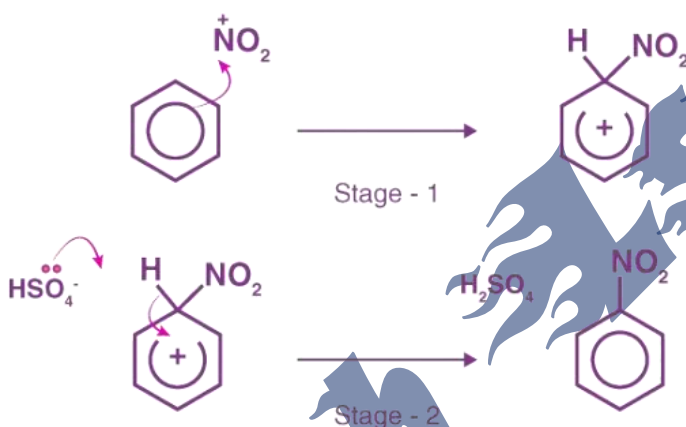


FIGURE 1 Production of nitrobenzene from benzene.

1. **Benzene Supply:** Start with benzene as the primary raw material.
2. **Nitrator:** Benzene is fed into a nitrator where it reacts with a mixture of concentrated nitric acid (HNO_3) and sulfuric acid (H_2SO_4) as a catalyst. This step forms nitrobenzene.

3. **Separator:** The reaction mixture is then transferred to a separator where the nitrobenzene is separated from the acid mixture.
4. **Washer:** The separated nitrobenzene is washed to remove any residual acids.
5. **Waste Water Treatment:** The wash water, containing traces of acids and other by-products, is sent to a waste water treatment unit.
6. **Nitrobenzene Collection:** The purified nitrobenzene is collected as the final product.

Reaction:



Q. What is the significance of ANNEALING in glass industry?

Repeat

Q. What are homogeneous CATALYSTS?

Homogeneous catalysts are catalysts that exist in the same phase as the reactants, typically in a liquid or gas state. They interact directly at a molecular level, leading to efficient and controlled reactions. These catalysts are widely used in organic synthesis, industrial processes, and biological systems like enzymatic reactions.

Example Reaction

Esterification Reaction:



Here, sulfuric acid (H_2SO_4) acts as a homogeneous catalyst in the reaction between acetic acid and methanol to form methyl acetate.

Advantages

1. **High Efficiency:** Since the catalyst is in the same phase as the reactants, it leads to a faster and more uniform reaction.
2. **Better Selectivity:** Homogeneous catalysts provide better control over the reaction, leading to fewer side products.

Disadvantages

1. **Difficult Separation:** Since the catalyst is mixed with the products, additional purification steps are needed.
2. **Shorter Catalyst Life:** Homogeneous catalysts may degrade or lose activity over time, requiring frequent replacement.

Q. What are Metallic soaps?

Repeat

Q. Describe any two SULFONATING agents?

1.2.2.2 Sulfonating agents

Following are some important sulfonating agents

i) Sulfur trioxide

Sulfur trioxide is the most efficient sulfonating agent as it involves the direct addition. It is principally used for sulfonating aromatic compounds (e.g. toluene, nitrobenzene, detergent alkylate), petroleum oils and long chain terminal alkanes.



ii) Sulfuric acid and Oleum

Sulfuric acid is the hydrate of sulfur trioxide and the most common reagent for direct sulfonation. Oleum is a solution of sulfur trioxide in 100% sulfuric acid.



iii) Chlorosulfonic acid (ClSO₃H)

It is an adduct of sulfur trioxide and hydrochloric acid is released as a gas during sulfonation. Excess of chlorosulfonic acid is used for sulfonyl chlorides. It is used for sulfating alcohols and to prepare sulfonyl chlorides.



Q. Why SO₃ is not directly added to the water in CONTAT process?

Repeat

Q. Describe the principle of multiple effect evaporator?

Repeat

Extensive Questions

Q. Describe manufacturing of ACETIC ACID in industry?

https://pureadmin.qub.ac.uk/ws/portalfiles/portal/202491264/Revised_clean_copy_.pdf

well explain

Q. What is a TEMPERED safety glass? How is it manufactured?

Repeat

Q. Describe SOAP manufacturing by KETTLE process.

Repeat

Q. How NITRATION of benzene is carried out in industry?

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1

[https://chem.libretexts.org/Bookshelves/Organic_Chemistry/Supplemental_Modules_\(Organic_Chemistry\)/Arenes/Reactivity_of_Arenes/Nitration_of_Benzene_and_Methylbenzene#:~:text=The%20nitration%20of%20benzene,-Nitration%20happens%20when&text=Benzen%20is%20treated%20with%20a,Yellow%20oily%20nitrobenzene%20is%20formed.](https://chem.libretexts.org/Bookshelves/Organic_Chemistry/Supplemental_Modules_(Organic_Chemistry)/Arenes/Reactivity_of_Arenes/Nitration_of_Benzene_and_Methylbenzene#:~:text=The%20nitration%20of%20benzene,-Nitration%20happens%20when&text=Benzen%20is%20treated%20with%20a,Yellow%20oily%20nitrobenzene%20is%20formed.)

2

<https://byjus.com/je/nitration/>

Q. Explain MULTIPLE EFFECT EVAPORATOR. Why is it preferred over single effect evaporator?

Repeat

**Beautiful Dua
from Quran**

رَبِّ اشْرَحْ لِي صَدْرِي وَيَسِّرْ لِي أَمْرِي
وَاحْلُلْ عُقْدَةً مِّن لِّسَانِي يَفْقَهُوا قَوْلِي

Punjab University

Bs (4 Years) Chemistry

Semester Sixth

Solve Past paper

(2018)

Desolate Dreamer
Desolate Dreamer
Desolate Dreamer

All repeat

**Beautiful Dua
from Quran**

رَبِّ اشْرَحْ لِي صَدْرِي وَيَسِّرْ لِي أَمْرِي
وَاحْلُلْ عُقْدَةً مِّن لِّسَانِي يَفْقَهُوا قَوْلِي

Punjab University

Bs (4 Years) Chemistry

Semester Sixth

Solve Past paper

(2017)

Desolate Dreamer
Desolate Dreamer
Desolate Dreamer

All Repeat